Finding North at Sea — compasses and magnet hunting ⏱ ~25 min

NAME _____ DUE _____ SCORE / EFFORT _____

1 · Read

Finding North at Sea

Before compasses, sailors found their way by carefully watching the Sun by day and the stars at night. On cloudy days and stormy nights, however, the sky could not help them, and ships could easily drift off course.

A magnetic compass changed this. Its needle is a small magnet that turns until one end points roughly north, so sailors could check their direction at any time, in almost any weather. Once they knew where north was, they could work out east, south and west too.

A ship's compass had to be looked after carefully. Big pieces of iron or steel nearby can pull the needle off course, so the compass was kept in a special stand, called a **binnacle**, near the ship's wheel. Many ship compasses float in liquid, which steadies the needle as the ship rolls on the waves.

Today, most ships also use satellite navigation. Even so, sailors still carry a compass, because it needs no batteries and keeps working when electronic equipment fails.

2 · Questions

1. How did sailors find their way **before** compasses? Why didn't this always work?

2. Why must a compass be kept away from big pieces of iron or steel?

3. Why do sailors still carry a compass today, even though ships have satellite navigation? Give **two** reasons from the text.

4. The text says *one end points roughly north*. Why do you think the writer used the word **roughly**?

3 · Adverb work

5. Underline the adverbial in each sentence and write whether it tells **how**, **when** or **where**.
a) *Sailors carefully watched the stars.* _____
b) *On cloudy days, the sky could not help them.* _____
c) *The compass was kept near the ship's wheel.* _____

6. Rewrite this sentence with the adverbial **fronted**, and add the comma:
The needle slowly settled after the storm.

7. *(Revision — Weeks 3 and 7)* Write **one compound sentence** about a compass that uses *but* or *so*, and includes a **possessive apostrophe** (e.g. *the sailor's…*).

4 · At home: magnet hunt 🧲

8. With an adult's permission, take a fridge magnet around your home. Gently test **eight** different objects. **Do not** test phones, computers, TVs, bank cards or anything electronic.
Predict first, then test, and record your results.

Object	What is it made of? (guess)	Prediction: stick? ✓/×	Result: stick? ✓/×

9. Which result **surprised** you most? Explain what you think the object is made of, using the words **attract** and a material name (e.g. *steel*, *aluminium*).

10. Teach someone at home **one** fact about magnets from this week. Write down who you taught and what you told them.